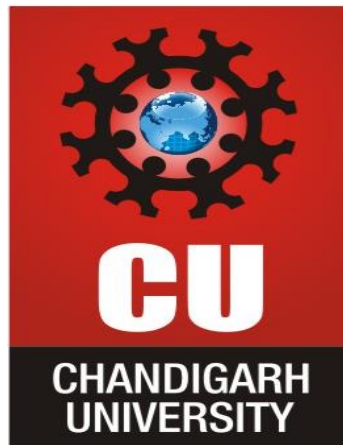




**HOMETASK:04**

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**SECTION** :24\_AIT\_KRG-G1-B

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**SUBJECT CODE** :24CSP-310

**SUBJECT NAME** :Advanced Database Management  
System

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[A].

## International Call Percentage

### Verizon SQL Interview Question

A phone call is considered an international call when the person calling is in a different country than the person receiving the call.

What percentage of phone calls are international? Round the result to 1 decimal.

Assumption:

- The caller\_id in phone\_info table refers to both the caller and receiver.

phone\_calls Table:

| Column Name | Type      |
|-------------|-----------|
| caller_id   | integer   |
| receiver_id | integer   |
| call_time   | timestamp |

phone\_calls Example Input:

| caller_id | receiver_id | call_time           |
|-----------|-------------|---------------------|
| 1         | 2           | 2022-07-04 10:13:49 |
| 1         | 5           | 2022-08-21 23:54:56 |
| 5         | 1           | 2022-05-13 17:24:06 |
| 5         | 6           | 2022-03-18 12:11:49 |

phone\_info Table:

| Column Name  | Type    |
|--------------|---------|
| caller_id    | integer |
| country_id   | integer |
| network      | integer |
| phone_number | string  |

phone\_info Example Input:

| caller_id | country_id | network  | phone_number    |
|-----------|------------|----------|-----------------|
| 1         | US         | Verizon  | +1-212-897-1964 |
| 2         | US         | Verizon  | +1-703-346-9529 |
| 3         | US         | Verizon  | +1-650-828-4774 |
| 4         | US         | Verizon  | +1-415-224-6663 |
| 5         | IN         | Vodafone | +91 7503-907302 |
| 6         | IN         | Vodafone | +91 2287-664895 |

Example Output:

| international_calls_pct |
|-------------------------|
| 50.0                    |

**SOLUTION:**

```
SELECT
    ROUND(
        100.0 * SUM(
            CASE
                WHEN caller.country_id <> receiver.country_id THEN 1
            ELSE 0
            END
        ) / COUNT(*),
        1
    ) AS international_calls_pct
FROM phone_calls pc
JOIN phone_info caller
ON pc.caller_id = caller.caller_id
JOIN phone_info receiver
ON pc.receiver_id = receiver.caller_id;
```